PAPER_568: Wavelength-Dependent Opacity κ_λ(λ) in the UQFF Olbers Framework

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

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§1 Abstract

The UQFF Olbers resolution in PAPER_564–566 is wavelength-independent: every photon receives the same DPM/VDS/BSFG suppression. In reality, dust and vacuum absorption depend strongly on photon wavelength — the night sky is not equally dark at all wavelengths. This paper introduces **wavelength-dependent opacity** $\kappa_\lambda(\lambda)$ into the 26-shell framework, yielding a **spectral Olbers resolution**: the sky is dark in the UV but slightly brighter in the far-IR. The UQFF [SSq] factor also acquires a spectral form.

$$B_n(\lambda) = \frac{n_\star L_\star(\lambda) \, \Delta r}{4\pi c (1+z_n)^4} \cdot e^{-\kappa_\lambda(\lambda_\text{em}) \, n_H \Delta r} \cdot e^{-[\text{SSq}](\lambda) \cdot n/26}$$

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§2 Classical Dust Opacity Power Law

Standard interstellar dust opacity:

$$\kappa_\lambda(\lambda) = \kappa_0 \left(\frac{\lambda}{\lambda_0}\right)^\beta$$

with $\beta \approx -2$ in the UV-optical regime (Draine 2003). Reference values:

Band	λ	$\kappa_\lambda / \kappa_V$
UV	100 nm	~ 10
V	550 nm	1.0 (reference)
NIR	2 µm	~ 0.03
FIR	100 µm	$\sim 2 \times 10^{-4}$

The universe is far more opaque in the UV than in the far-IR — explaining why the UV night sky is darker than the faint IR glow.

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§3 UQFF Spectral [SSq]

Within UQFF, the 26-dimensional vacuum coupling $[\text{SSq}]$ acquires a spectral form via the VDS photon-wavelength modulation (PAPER_429):

$$[\text{SSq}](\lambda) = [\text{SSq}]_0 \cdot \left(\frac{\lambda_\text{opt}}{\lambda}\right)^{1/26}$$

with $\lambda_\text{opt} = 550$ nm and $[\text{SSq}]_0 = 0.507$. This gives:

Band	λ	$[\text{SSq}](\lambda)$
UV (100 nm)	UV	$0.507 \times (550/100)^{1/26} \approx 0.574$
V (550 nm)	Optical	0.507
NIR (2 µm)	NIR	$0.507 \times (0.55/2.0)^{1/26} \approx 0.460$
FIR (100 µm)	FIR	$0.507 \times (0.55/100)^{1/26} \approx 0.370$

The UV receives *stronger* [SSq] suppression; FIR receives *weaker* suppression — consistent with the IR EBL being the dominant contribution to $B_\text{sky,obs}$.

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§4 Spectral Shell Brightness

Combined spectral opacity and VDS suppression per shell:

$$B_n(\lambda) = \frac{n_\star(\lambda) L_\star(\lambda) \Delta r}{4\pi c (1+z_n)^4} \cdot e^{-\kappa_\lambda(\lambda_\text{em}) n_H \Delta r} \cdot e^{-[\text{SSq}](\lambda) \cdot n/26}$$

where $\lambda_\text{em} = \lambda_\text{obs}(1+z_n)$ (blueshifted emitted wavelength), $n_H = 1.6 \times 10^{-7}$ m^{-3} (mean hydrogen column density).

Integrated sky brightness per wavelength:

$$B_\text{sky}(\lambda) = \sum_{n=1}^{26} B_n(\lambda)$$

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§5 Spectral Olbers Prediction

Integrating over all wavelengths:

$$B_\text{sky}^\text{total} = \int_0^\infty B_\text{sky}(\lambda) \, d\lambda$$

Key spectral features:

- **UV ($\lambda < 200$ nm):** Strong opacity + enhanced [SSq] → nearly zero contribution
- **Optical ($\lambda \approx 550$ nm):** Standard [SSq] = 0.507 suppression (PAPER_564)
- **NIR/FIR ($\lambda > 1$ μm):** Reduced opacity + reduced [SSq] → dominant contributor to EBL

This explains the observational fact (EBL measurements) that the extragalactic background light peaks in the far-IR (~ 100–140 μm) and optical.

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§6 Numerical Estimates

For the optical band ($\lambda = 550$ nm, $n_H \Delta r = $ column depth):

$$\tau_V = \kappa_V n_H \Delta r \approx 10^{-7} \, \text{per shell} \quad \text{(intergalactic medium is highly transparent)}$$

The dust opacity is negligible in the intergalactic medium; the dominant suppression comes from VDS [SSq] and the DPM cascade. The wavelength-dependent correction is order $(\lambda_\text{opt}/\lambda)^{1/26}$, giving a $\pm 20\%$ correction across the full optical range.

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§7 Testable Predictions

1. **IR excess:** B_sky peaks at $\lambda \sim 100$ μm (FIR) due to reduced [SSq] suppression — consistent with COBE/Herschel EBL measurements.
2. **UV opacity edge:** $B_\text{sky}(\text{UV})$ should be suppressed by a factor $\sim e^{-[\text{SSq}]_\text{UV} \cdot 26 / 2}$ relative to optical — testable with GALEX UV background.
3. **[SSq] spectral slope:** The spectral index $d \ln [\text{SSq}] / d \ln \lambda = -1/26$ — a unique UQFF prediction (compared to $\beta = -2$ dust law).

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§8 Builds On / Addresses

Paper	Role
PAPER_564	DPM 26-shell baseline (wavelength-independent — extended here)
PAPER_565	VDS [SSq] value (acquires spectral form here)
PAPER_566	Gap analysis — this is Missing Extension 2

PAPER_568 — Star Magic UQFF Framework — QS 5/5